# VIT BHOPAL UNIVERSITY, MP

Course Name : Probability, Statistics and Reliability
Course Code : MAT3003
Practice question CAT2

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**Q I.** An engineering team is testing a two-component dual-stage cooling system. Let $X$ and $Y$ denote the normalized operating lifetimes of the primary and auxiliary cooling units, respectively. Their joint probability density function is modeled as:

$$f(x,y) = \begin{cases} Kx(x+y^2), & 0 < x < 1,\ 0 < y < 3-2x \\ 0, & \text{otherwise} \end{cases}$$

**(1)** Find the value of $K$.

**(2)** The probability that the auxiliary unit operates for more than 1 unit, given that the primary unit has an operating lifetime of $\frac{1}{2}$ unit.

**(3)** The probability that the combined lifetime exceeds 2 units.

**Q II.** A sample of 4900 male athletes from Region $A$ has a mean height of 172.4 cm with a standard deviation of 6.2 cm. Another independent sample of 2500 male athletes from Region $B$ has a mean height of 173.8 cm with a standard deviation of 6.0 cm. State the null and alternative hypotheses, and test at the 1% level of significance whether the data indicates that athletes from Region $B$ are, on average, taller than athletes from Region $A$.

**Q III.** An engineering team is evaluating the failure times of two critical sensors, $X$ and $Y$ (in thousands of hours), operating inside a high-temperature reactor. Their joint probability density function is given by:

$$f(x,y) = \begin{cases} c(x+2y), & 0 < x < 1,\ 0 < y < 1-x^2 \\ 0, & \text{otherwise} \end{cases}$$

**(1)** Determine the value of the constant $c$.

**(2)** Find the conditional probability $P(Y > 0.5 \mid X = 0.25)$.

**(3)** Find the probability that the combined lifetime $X+Y$ exceeds 0.5 units.

**Q.IV** An agricultural research station conducted a field study to evaluate the effectiveness of soil moisture retention under varying solar radiation levels. Field observations were recorded across 8 independent sampling intervals to analyze the relationship between daily solar radiation exposure ($X$) and the corresponding soil moisture loss ($Y$):

| $X$ | 3.5 | 4.2 | 5.1 | 5.8 | 6.4 | 4.8 | 3.1 | 2.5 |
|---|---|---|---|---|---|---|---|---|
| $Y$ | 10.2 | 11.5 | 12.8 | 13.4 | 14.1 | 12.0 | 9.8 | 8.6 |

Find the lines of regression of $Y$ on $X$ and $X$ on $Y$, and calculate the Karl Pearson correlation coefficient between solar radiation exposure and soil moisture loss.

**Q.V** A chemical process laboratory investigated the relationship between reaction temperature and the yield of a refined compound. Experimental measurements were collected across 8 trials

to evaluate the functional dependency of reaction temperature ($X$, in °C) on the mass of unreacted byproduct ($Y$, in grams):

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| $X$ | 12.5 | 14.0 | 15.8 | 16.5 | 18.2 | 13.6 | 11.0 | 10.4 |
| $Y$ | 25.4 | 23.1 | 20.5 | 19.2 | 17.0 | 24.0 | 27.2 | 28.5 |

Find the equations of both regression lines and determine the correlation coefficient between the reaction temperature and the quantity of unreacted byproduct.

**Q VI.** Suppose two independent sensor networks register daily event counts represented by discrete random variables $X$ and $Y$. The joint probability mass function governing these systems is defined as follows: the random variable $X$ takes values in $\{1,2\}$, while $Y$ takes values in $\{0,1,2\}$. The joint probability mass function of two discrete random variables $X$ and $Y$ is given by the following tabular distribution:

| $f(x,y)$ | $Y=0$ | $Y=1$ | $Y=2$ |
|---|---|---|---|
| $X=1$ | 0.10 | 0.15 | 0.05 |
| $X=2$ | 0.20 | 0.35 | 0.15 |

Based on this joint distribution, determine:

**(1)** The marginal probability mass functions $f_X(x)$ and $f_Y(y)$.

**(2)** The conditional distribution of $Y$ given $X=2$, denoted as $P(Y=y \mid X=2)$.

**(3)** The expected value $E[XY]$.

**(4)** Find $P(X+Y>2)$.

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